# Safety data sheet

according to Regulation (EC) No. 1907/2006 (REACH), amended by 2015/830/EU

**m-Xylene ≥ 99%, for synthesis**

article number: **3791**
Version: **2.0 en**
Replaces version of: 2016-07-04
Version: (1)

date of compilation: 2016-06-29
Revision: 2019-03-08

## SECTION 1: Identification of the substance/mixture and of the company/ undertaking

### 1.1 Product identifier

| | |
|---|---|
| Identification of the substance | **m-Xylene** |
| Article number | 3791 |
| Registration number (REACH) | It is not required to list the identified uses because the substance is not subject to registration according to REACH (< 1 t/a) |
| Index No | 601-022-00-9 |
| EC number | 203-576-3 |
| CAS number | 108-38-3 |

### 1.2 Relevant identified uses of the substance or mixture and uses advised against

| | |
|---|---|
| **Identified uses:** | laboratory chemical<br>laboratory and analytical use |

### 1.3 Details of the supplier of the safety data sheet

Carl Roth GmbH + Co KG
Schoemperlenstr. 3-5
D-76185 Karlsruhe
Germany

**Telephone:** +49 (0) 721 - 56 06 0
**Telefax:** +49 (0) 721 - 56 06 149
**e-mail:** sicherheit@carlroth.de
**Website:** www.carlroth.de

| | |
|---|---|
| Competent person responsible for the safety data sheet | : Department Health, Safety and Environment |
| **e-mail (competent person)** | **: sicherheit@carlroth.de** |

### 1.4 Emergency telephone number

| | |
|---|---|
| Emergency information service | **Poison Centre Munich: +49/(0)89 19240** |

## SECTION 2: Hazards identification

### 2.1 Classification of the substance or mixture

**Classification according to Regulation (EC) No 1272/2008 (CLP)**

**Classification acc. to GHS**

| Section | Hazard class | Hazard class and category | Hazard statement |
|---|---|---|---|
| 2.6 | flammable liquid | (Flam. Liq. 3) | H226 |
| 3.1D | acute toxicity (dermal) | (Acute Tox. 4) | H312 |
| 3.1I | acute toxicity (inhal.) | (Acute Tox. 4) | H332 |
| 3.2 | skin corrosion/irritation | (Skin Irrit. 2) | H315 |

| Classification acc. to GHS | | | |
|---|---|---|---|
| **Section** | **Hazard class** | **Hazard class and category** | **Hazard statement** |
| 3.3 | serious eye damage/eye irritation | (Eye Irrit. 2) | H319 |
| 3.8R | specific target organ toxicity - single exposure (respiratory tract irritation) | (STOT SE 3) | H335 |
| 3.9 | specific target organ toxicity - repeated exposure | (STOT RE 2) | H373 |
| 3.10 | aspiration hazard | (Asp. Tox. 1) | H304 |

## 2.2 Label elements

### Labelling according to Regulation (EC) No 1272/2008 (CLP)

**Signal word** **Danger**

**Pictograms**

GHS02, GHS07, GHS08



**Hazard statements**

| | |
|---|---|
| H226 | Flammable liquid and vapour |
| H304 | May be fatal if swallowed and enters airways |
| H312+H332 | Harmful in contact with skin or if inhaled |
| H315 | Causes skin irritation |
| H319 | Causes serious eye irritation |
| H335 | May cause respiratory irritation |
| H373 | May cause damage to organs (respiratory system, nervous system) through prolonged or repeated exposure |

**Precautionary statements**

**Precautionary statements - prevention**

| | |
|---|---|
| P210 | Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. |
| P260 | Do not breathe mist/vapours/spray. |
| P280 | Wear protective gloves/protective clothing/eye protection/face protection. |

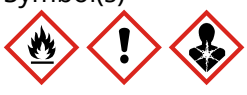
**Precautionary statements - response**

| | |
|---|---|
| P301+P310 | IF SWALLOWED: Immediately call a POISON CENTER/doctor/... |
| P302+P352 | IF ON SKIN: Wash with plenty of soap and water. |
| P305+P351+P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P331 | Do NOT induce vomiting. |

**Labelling of packages where the contents do not exceed 125 ml**

Signal word: **Danger**

Symbol(s)



| | |
|---|---|
| H304 | May be fatal if swallowed and enters airways. |
| P301+P310 | IF SWALLOWED: Immediately call a POISON CENTER/doctor. |
| P331 | Do NOT induce vomiting. |

### 2.3 Other hazards

There is no additional information.

## SECTION 3: Composition/information on ingredients

### 3.1 Substances

| | |
|---|---|
| Name of substance | 1,3-Dimethylbenzene |
| Index No | 601-022-00-9 |
| EC number | 203-576-3 |
| CAS number | 108-38-3 |
| Molecular formula | $C_8H_{10}$ |
| Molar mass | 106,2 $^{g}/_{mol}$ |

## SECTION 4: First aid measures

### 4.1 Description of first aid measures



**General notes**

Take off contaminated clothing.

**Following inhalation**

Provide fresh air. In all cases of doubt, or when symptoms persist, seek medical advice.

**Following skin contact**

Rinse skin with water/shower. In case of skin irritation, consult a physician.

**Following eye contact**

Irrigate copiously with clean, fresh water for at least 10 minutes, holding the eyelids apart. In case of eye irritation consult an ophthalmologist.

**Following ingestion**

Rinse mouth. Do not induce vomiting. Aspiration hazard. Call a physician immediately.

### 4.2 Most important symptoms and effects, both acute and delayed

Irritation, Cough, Headache, Impairment of vision, Dizziness, Vertigo, Nausea, Vomiting, Diarrhoea, Breathing difficulties, Unconsciousness, Aspiration hazard

### 4.3 Indication of any immediate medical attention and special treatment needed

none

## SECTION 5: Firefighting measures

### 5.1 Extinguishing media



**Suitable extinguishing media**

Co-ordinate fire-fighting measures to the fire surroundings
water spray, foam, dry extinguishing powder, carbon dioxide (CO2)

**Unsuitable extinguishing media**

water jet

### 5.2 Special hazards arising from the substance or mixture

Combustible. Vapours are heavier than air, spread along floors and form explosive mixtures with air.

**Hazardous combustion products**

In case of fire may be liberated: carbon monoxide (CO), carbon dioxide (CO2)

### 5.3 Advice for firefighters

Vapours are heavier than air. Beware of reignition. Fight fire with normal precautions from a reasonable distance. Wear self-contained breathing apparatus.

## SECTION 6: Accidental release measures

### 6.1 Personal precautions, protective equipment and emergency procedures



**For non-emergency personnel**

Use personal protective equipment as required. Avoid contact with skin, eyes and clothes. Do not breathe vapour/spray. Avoidance of ignition sources.

### 6.2 Environmental precautions

Keep away from drains, surface and ground water. Explosive properties.

### 6.3 Methods and material for containment and cleaning up

**Advices on how to contain a spill**

Covering of drains.

**Advices on how to clean up a spill**

Absorb with liquid-binding material (e.g. sand, diatomaceous earth, acid- or universal binding agents).

**Other information relating to spills and releases**

Place in appropriate containers for disposal. Ventilate affected area.

### 6.4 Reference to other sections

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

## SECTION 7: Handling and storage

### 7.1 Precautions for safe handling

Provide adequate ventilation as well as local exhaustion at critical locations. Avoid exposure. When not in use, keep containers tightly closed.

**• Measures to prevent fire as well as aerosol and dust generation**


Keep away from sources of ignition - No smoking.

Take precautionary measures against static discharge.

**Advice on general occupational hygiene**

Wash hands before breaks and after work. Keep away from food, drink and animal feedingstuffs. When using do not smoke.

### 7.2 Conditions for safe storage, including any incompatibilities

Store in a well-ventilated place. Keep container tightly closed. Protect from sunlight.

**Incompatible substances or mixtures**

Observe hints for combined storage.

**Consideration of other advice**

Ground/bond container and receiving equipment.

**• Ventilation requirements**

Use local and general ventilation.

**• Specific designs for storage rooms or vessels**

Recommended storage temperature: 15 – 25 °C.

### 7.3 Specific end use(s)

No information available.

## SECTION 8: Exposure controls/personal protection

### 8.1 Control parameters

**National limit values**

**Occupational exposure limit values (Workplace Exposure Limits)**

| Country | Name of agent | CAS No | Identifier | TWA [ppm] | TWA [mg/m³] | STEL [ppm] | STEL [mg/m³] | Source |
|---|---|---|---|---|---|---|---|---|
| EU | m-xylene | 108-38-3 | IOELV | 50 | 221 | 100 | 442 | 2000/39/EC |
| GB | m-xylene | 108-38-3 | WEL | 50 | 220 | 100 | 441 | EH40/2005 |

**Notation**

STEL Short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15-minute period (unless otherwise specified)

TWA Time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours time-weighted average (unless otherwise specified)

### Biological limit values

| Coun-try | Name of agent | Parameter | Identifier | Value | Material | Source |
|---|---|---|---|---|---|---|
| GB | m-xylene | methylhippuric acids | BMGV | 650 mmol/mol | urine | EH40/2005 |

### Relevant DNELs/DMELs/PNECs and other threshold levels

**• human health values**

| Endpoint | Threshold level | Protection goal, route of exposure | Used in | Exposure time |
|---|---|---|---|---|
| DNEL | 221 mg/m³ | human, inhalatory | worker (industry) | chronic - systemic effects |
| DNEL | 442 mg/m³ | human, inhalatory | worker (industry) | acute - systemic effects |
| DNEL | 221 mg/m³ | human, inhalatory | worker (industry) | chronic - local effects |
| DNEL | 442 mg/m³ | human, inhalatory | worker (industry) | acute - local effects |
| DNEL | 212 mg/kg bw/day | human, dermal | worker (industry) | chronic - systemic effects |

**• environmental values**

| Endpoint | Threshold level | Environmental compartment |
|---|---|---|
| PNEC | 0,25 $^{mg}/_{l}$ | water |
| PNEC | 0,044 $^{mg}/_{l}$ | freshwater |
| PNEC | 0,004 $^{mg}/_{l}$ | marine water |
| PNEC | 1,6 $^{mg}/_{l}$ | sewage treatment plant (STP) |
| PNEC | 2,52 $^{mg}/_{kg}$ | freshwater sediment |
| PNEC | 0,252 $^{mg}/_{kg}$ | marine sediment |
| PNEC | 0,852 $^{mg}/_{kg}$ | soil |

## 8.2 Exposure controls

### Individual protection measures (personal protective equipment)

### Eye/face protection



Use safety goggle with side protection.

### Skin protection



**• hand protection**

Wear suitable gloves. Chemical protection gloves are suitable, which are tested according to EN 374. For special purposes, it is recommended to check the resistance to chemicals of the protective gloves mentioned above together with the supplier of these gloves.

- **type of material**

FKM (fluoro rubber)

- **material thickness**

0,4 mm.

- **breakthrough times of the glove material**

>480 minutes (permeation: level 6)

- **other protection measures**

Take recovery periods for skin regeneration. Preventive skin protection (barrier creams/ointments) is recommended.

**Respiratory protection**



Respiratory protection necessary at: Aerosol or mist formation. Type: A (against organic gases and vapours with a boiling point of > 65 °C , colour code: Brown).

**Environmental exposure controls**

Keep away from drains, surface and ground water.

## SECTION 9: Physical and chemical properties

### 9.1 Information on basic physical and chemical properties

**Appearance**

| | |
|---|---|
| Physical state | liquid (fluid) |
| Colour | colourless |
| Odour | characteristic |
| Odour threshold | No data available |

**Other physical and chemical parameters**

| | |
|---|---|
| pH (value) | This information is not available. |
| Melting point/freezing point | -47,8 °C at 1.013 hPa |
| Initial boiling point and boiling range | 139,1 °C at 1.013 hPa |
| Flash point | 27 °C at 1.013 hPa |
| Evaporation rate | no data available |
| Flammability (solid, gas) | not relevant (fluid) |
| Explosive limits | |
| • lower explosion limit (LEL) | 1,1 vol% |
| • upper explosion limit (UEL) | 7 vol% |
| Explosion limits of dust clouds | not relevant |
| Vapour pressure | 8 hPa at 20 °C |
| Density | 0,86 $^{g}/_{cm^3}$ at 25 °C |
| Vapour density | 3,66 (air = 1) |

| | |
|---|---|
| Bulk density | Not applicable |
| Relative density | Information on this property is not available. |
| Solubility(ies) | |
| Water solubility | ~ 146 $^{mg}/_{l}$ at 25 °C |
| Partition coefficient | |
| n-octanol/water (log KOW) | 3,2 (pH value: 7, 20 °C) (ECHA) |
| Soil organic carbon/water (log KOC) | 2,73 (ECHA) |
| Auto-ignition temperature | 528 °C at 1.013 hPa - ECHA |
| Decomposition temperature | no data available |
| Viscosity | |
| • kinematic viscosity | 0,6756 $^{mm^2}/_{s}$ |
| • dynamic viscosity | 0,581 mPa s at 25 °C |
| Explosive properties | Shall not be classified as explosive |
| Oxidising properties | none |

**9.2 Other information**

| | |
|---|---|
| Surface tension | 28,01 $^{mN}/_{m}$ (25 °C) |
| Temperature class (EU, acc. to ATEX) | T1 (Maximum permissible surface temperature on the equipment: 450°C) |

## SECTION 10: Stability and reactivity

**10.1 Reactivity**

Risk of ignition. In case of warming: Vapours can form explosive mixtures with air.

**10.2 Chemical stability**

The material is stable under normal ambient and anticipated storage and handling conditions of temperature and pressure.

**10.3 Possibility of hazardous reactions**

Violent reaction with: Oxidisers, Nitric acid, Sulphuric acid, Sulphur, Acids

**10.4 Conditions to avoid**

Keep away from heat.

**10.5 Incompatible materials**

plastic and rubber

**10.6 Hazardous decomposition products**

Hazardous combustion products: see section 5.

# SECTION 11: Toxicological information

## 11.1 Information on toxicological effects

### Acute toxicity

| Exposure route | Endpoint | Value | Species | Source |
|---|---|---|---|---|
| oral | LD50 | 3.523 $^{mg}/_{kg}$ | rat | ECHA |

### Skin corrosion/irritation

Causes skin irritation.

### Serious eye damage/eye irritation

Causes serious eye irritation.

### Respiratory or skin sensitisation

Shall not be classified as a respiratory or skin sensitiser.

### Summary of evaluation of the CMR properties

Shall not be classified as germ cell mutagenic, carcinogenic nor as a reproductive toxicant

**• Specific target organ toxicity - single exposure**

May cause respiratory irritation.

**• Specific target organ toxicity - repeated exposure**

May cause damage to organs (respiratory system, nervous system) through prolonged or repeated exposure.

### Aspiration hazard

May be fatal if swallowed and enters airways.

### Symptoms related to the physical, chemical and toxicological characteristics

**• If swallowed**

diarrhoea, vomiting, aspiration hazard

**• If in eyes**

Causes serious eye irritation

**• If inhaled**

irritant effects, cough, breathing difficulties, pulmonary oedema

**• If on skin**

causes skin irritation, risk of absorption via the skin

### Other information

Other adverse effects: Headache, Impairment of vision, Dizziness, Vertigo, Nausea, Dyspnoea, Unconsciousness, Liver and kidney damage, Symptoms can occur only after several hours

## SECTION 12: Ecological information

### 12.1 Toxicity

acc. to 1272/2008/EC: Shall not be classified as hazardous to the aquatic environment.

**Aquatic toxicity (acute)**

| Endpoint | Value | Species | Source | Exposure time |
|---|---|---|---|---|
| LC50 | 2,6 $^{mg}/_{l}$ | rainbow trout | ECHA | 96 h |
| ErC50 | 4,7 $^{mg}/_{l}$ | algae | ECHA | 72 h |

**Aquatic toxicity (chronic)**

| Endpoint | Value | Species | Source | Exposure time |
|---|---|---|---|---|
| EC50 | 2,2 $^{mg}/_{l}$ | algae | ECHA | 73 h |
| NOEC | 0,714 $^{mg}/_{l}$ | striped brill | ECHA | 35 d |
| NOEC | 1,57 $^{mg}/_{l}$ | aquatic invertebrates | ECHA | 21 d |
| NOEC | 0,44 $^{mg}/_{l}$ | algae | ECHA | 73 h |

### 12.2 Process of degradability

The substance is readily biodegradable.
Theoretical Oxygen Demand: 3,165 $^{mg}/_{mg}$
Theoretical Carbon Dioxide: 3,316 $^{mg}/_{mg}$

| Process | Degradation rate | Time |
|---|---|---|
| oxygen depletion | 90 % | 28 d |

### 12.3 Bioaccumulative potential

Does not significantly accumulate in organisms.

| | |
|---|---|
| n-octanol/water (log KOW) | 3,2 (pH value: 7, 20 °C) |
| BCF | >5,5 – <12,2 |

### 12.4 Mobility in soil

| | |
|---|---|
| Henry's law constant | 623 $^{Pa\ m^3}/_{mol}$ at 25 °C |
| The Organic Carbon normalised adsorption coefficient | 2,73 |

### 12.5 Results of PBT and vPvB assessment

Data are not available.

### 12.6 Other adverse effects

Data are not available.

## SECTION 13: Disposal considerations

### 13.1 Waste treatment methods



This material and its container must be disposed of as hazardous waste. Dispose of contents/container in accordance with local/regional/national/international regulations.

**Sewage disposal-relevant information**

Do not empty into drains.

**Waste treatment of containers/packagings**

It is a dangerous waste; only packagings which are approved (e.g. acc. to ADR) may be used.

**Sewage disposal-relevant information**

Do not empty into drains.

**Waste treatment of containers/packagings**

It is a dangerous waste; only packagings which are approved (e.g. acc. to ADR) may be used.

### 13.2 Relevant provisions relating to waste

The allocation of waste identity numbers/waste descriptions must be carried out according to the EEC, specific to the industry and process.

### 13.3 Remarks

Waste shall be separated into the categories that can be handled separately by the local or national waste management facilities. Please consider the relevant national or regional provisions.

## SECTION 14: Transport information

| | | |
|---|---|---|
| **14.1** | UN number | **1307** |
| **14.2** | UN proper shipping name | **XYLENES** |
| | Hazardous ingredients | m-Xylene |
| **14.3** | Transport hazard class(es) |  |
| | Class | 3 (flammable liquids) |
| **14.4** | Packing group | III (substance presenting low danger) |
| **14.5** | Environmental hazards | none (non-environmentally hazardous acc. to the dangerous goods regulations) |

### 14.6 Special precautions for user

Provisions for dangerous goods (ADR) should be complied within the premises.

### 14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code

The cargo is not intended to be carried in bulk.

### 14.8 Information for each of the UN Model Regulations

**• Transport of dangerous goods by road, rail and inland waterway (ADR/RID/ADN)**

| | |
|---|---|
| UN number | 1307 |

| | |
|---|---|
| Proper shipping name | XYLENES |
| Particulars in the transport document | UN1307, XYLENES, 3, III, (D/E) |
| Class | 3 |
| Classification code | F1 |
| Packing group | III |
| Danger label(s) | 3 |



| | |
|---|---|
| Excepted quantities (EQ) | E1 |
| Limited quantities (LQ) | 5 L |
| Transport category (TC) | 3 |
| Tunnel restriction code (TRC) | D/E |
| Hazard identification No | 30 |
| **Emergency Action Code** | 3YE |
| **• International Maritime Dangerous Goods Code (IMDG)** | |
| UN number | 1307 |
| Proper shipping name | XYLENES |
| Particulars in the shipper's declaration | UN1307, XYLENES, 3, III, 27°C c.c. |
| Class | 3 |
| Marine pollutant | - |
| Packing group | III |
| Danger label(s) | 3 |



| | |
|---|---|
| Special provisions (SP) | 223 |
| Excepted quantities (EQ) | E1 |
| Limited quantities (LQ) | 5 L |
| EmS | F-E, S-D |
| Stowage category | A |
| **• International Civil Aviation Organization (ICAO-IATA/DGR)** | |
| UN number | 1307 |
| Proper shipping name | Xylenes |
| Particulars in the shipper's declaration | UN1307, Xylenes, 3, III |
| Class | 3 |
| Packing group | III |

| | |
|---|---|
| Danger label(s) | 3 |



| | |
|---|---|
| Special provisions (SP) | A3 |
| Excepted quantities (EQ) | E1 |
| Limited quantities (LQ) | 10 L |

## SECTION 15: Regulatory information

### 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

**Relevant provisions of the European Union (EU)**

**• Regulation 649/2012/EU concerning the export and import of hazardous chemicals (PIC)**

Not listed.

**• Regulation 1005/2009/EC on substances that deplete the ozone layer (ODS)**

Not listed.

**• Regulation 850/2004/EC on persistent organic pollutants (POP)**

Not listed.

**• Restrictions according to REACH, Annex XVII**

| Name of substance | CAS No | Wt% | Type of registration | Conditions of restriction | No |
|---|---|---|---|---|---|
| m-Xylene | | 100 | 1907/2006/EC annex XVII | R3 | 3 |
| m-Xylene | | 100 | 1907/2006/EC annex XVII | R40 | 40 |

**Legend**

R3 1. Shall not be used in:
- ornamental articles intended to produce light or colour effects by means of different phases, for example in ornamental lamps and ashtrays,
- tricks and jokes,
- games for one or more participants, or any article intended to be used as such, even with ornamental aspects,
2. Articles not complying with paragraph 1 shall not be placed on the market.
3. Shall not be placed on the market if they contain a colouring agent, unless required for fiscal reasons, or perfume, or both, if they:
- can be used as fuel in decorative oil lamps for supply to the general public, and,
- present an aspiration hazard and are labelled with R65 or H304,
4. Decorative oil lamps for supply to the general public shall not be placed on the market unless they conform to the European Standard on Decorative oil lamps (EN 14059) adopted by the European Committee for Standardisation (CEN).
5. Without prejudice to the implementation of other Community provisions relating to the classification, packaging and labelling of dangerous substances and mixtures, suppliers shall ensure, before the placing on the market, that the following requirements are met:
(a) lamp oils, labelled with R65 or H304, intended for supply to the general public are visibly, legibly and indelibly marked as follows: 'Keep lamps filled with this liquid out of the reach of children'; and, by 1 December 2010, 'Just a sip of lamp oil - or even sucking the wick of lamps - may lead to life-threatening lung damage';
(b) grill lighter fluids, labelled with R65 or H304, intended for supply to the general public are legibly and indelibly marked by 1 December 2010 as follows: 'Just a sip of grill lighter may lead to life threatening lung damage';
(c) lamp oils and grill lighters, labelled with R65 or H304, intended for supply to the general public are packaged in black opaque containers not exceeding 1 litre by 1 December 2010.
6. No later than 1 June 2014, the Commission shall request the European Chemicals Agency to prepare a dossier, in accordance with Article 69 of the present Regulation with a view to ban, if appropriate, grill lighter fluids and fuel for decorative lamps, labelled R65 or H304, intended for supply to the general public.
7. Natural or legal persons placing on the market for the first time lamp oils and grill lighter fluids, labelled with R65 or H304, shall by 1 December 2011, and annually thereafter, provide data on alternatives to lamp oils and grill lighter fluids labelled R65 or H304 to the competent authority in the Member State concerned. Member States shall make those data available to the Commission.

**Legend**

R40 1. Shall not be used, as substance or as mixtures in aerosol dispensers where these aerosol dispensers are intended for supply to the general public for entertainment and decorative purposes such as the following:
- metallic glitter intended mainly for decoration,
- artificial snow and frost,
- 'whoopee' cushions,
- silly string aerosols,
- imitation excrement,
- horns for parties,
- decorative flakes and foams,
- artificial cobwebs,
- stink bombs.

2. Without prejudice to the application of other Community provisions on the classification, packaging and labelling of substances, suppliers shall ensure before the placing on the market that the packaging of aerosol dispensers referred to above is marked visibly, legibly and indelibly with:
'For professional users only'.
3. By way of derogation, paragraphs 1 and 2 shall not apply to the aerosol dispensers referred to Article 8 (1a) of Council Directive 75/324/EEC (2).
4. The aerosol dispensers referred to in paragraphs 1 and 2 shall not be placed on the market unless they conform to the requirements indicated.

**• Restrictions according to REACH, Title VIII**

None.

**• List of substances subject to authorisation (REACH, Annex XIV)/SVHC - candidate list**

not listed

**• Seveso Directive**

**2012/18/EU (Seveso III)**

| No | Dangerous substance/hazard categories | Qualifying quantity (tonnes) for the application of lower and upper-tier requirements | | Notes |
|---|---|---|---|---|
| P5c | flammable liquids (cat. 2, 3) | 5.000 | 50.000 | 51) |

**Notation**

51) Flammable liquids, categories 2 or 3 not covered by P5a and P5b

**• Directive 75/324/EEC relating to aerosol dispensers**

**Filling batch**

**Deco-Paint Directive (2004/42/EC)**

| | |
|---|---|
| **VOC content** | 100 %<br>860 $^{g}/_{l}$ |

**Directive on industrial emissions (VOCs, 2010/75/EU)**

| | |
|---|---|
| **VOC content** | 100 % |
| **VOC content** | 860 $^{g}/_{l}$ |

**Directive 2011/65/EU on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS) - Annex II**

not listed

**Regulation 166/2006/EC concerning the establishment of a European Pollutant Release and Transfer Register (PRTR)**

not listed

**Directive 2000/60/EC establishing a framework for Community action in the field of water policy (WFD)**

not listed

### Regulation 98/2013/EU on the marketing and use of explosives precursors

not listed

### Regulation 111/2005/EC laying down rules for the monitoring of trade between the Community and third countries in drug precursors

not listed

### National inventories

Substance is listed in the following national inventories:

| Country | National inventories | Status |
|---|---|---|
| AU | AICS | substance is listed |
| CA | DSL | substance is listed |
| CN | IECSC | substance is listed |
| EU | ECSI | substance is listed |
| EU | REACH Reg. | substance is listed |
| JP | CSCL-ENCS | substance is listed |
| JP | ISHA-ENCS | substance is listed |
| KR | KECI | substance is listed |
| MX | INSQ | substance is listed |
| NZ | NZIoC | substance is listed |
| PH | PICCS | substance is listed |
| TW | TCSI | substance is listed |
| US | TSCA | substance is listed |

**Legend**

| | |
|---|---|
| AICS | Australian Inventory of Chemical Substances |
| CSCL-ENCS | List of Existing and New Chemical Substances (CSCL-ENCS) |
| DSL | Domestic Substances List (DSL) |
| ECSI | EC Substance Inventory (EINECS, ELINCS, NLP) |
| IECSC | Inventory of Existing Chemical Substances Produced or Imported in China |
| INSQ | National Inventory of Chemical Substances |
| ISHA-ENCS | Inventory of Existing and New Chemical Substances (ISHA-ENCS) |
| KECI | Korea Existing Chemicals Inventory |
| NZIoC | New Zealand Inventory of Chemicals |
| PICCS | Philippine Inventory of Chemicals and Chemical Substances |
| REACH Reg. | REACH registered substances |
| TCSI | Taiwan Chemical Substance Inventory |
| TSCA | Toxic Substance Control Act |

## 15.2 Chemical Safety Assessment

No Chemical Safety Assessment has been carried out for this substance.

# SECTION 16: Other information

### Abbreviations and acronyms

| Abbr. | Descriptions of used abbreviations |
|---|---|
| 2000/39/EC | Comission Directive establishing a first list of indicative occupational exposure limit values in implementation of Council Directive 98/24/EC |
| ADN | Accord européen relatif au transport international des marchandises dangereuses par voies de navigation intérieures (European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways) |
| ADR | Accord européen relatif au transport international des marchandises dangereuses par route (European Agreement concerning the International Carriage of Dangerous Goods by Road) |
| BCF | bioconcentration factor |
| CAS | Chemical Abstracts Service (service that maintains the most comprehensive list of chemical substances) |
| CLP | Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures |
| CMR | Carcinogenic, Mutagenic or toxic for Reproduction |
| DGR | Dangerous Goods Regulations (see IATA/DGR) |
| DMEL | Derived Minimal Effect Level |
| DNEL | Derived No-Effect Level |
| EH40/2005 | EH40/2005 Workplace exposure limits (http://www.nationalarchives.gov.uk/doc/open-government-licence/) |
| EINECS | European Inventory of Existing Commercial Chemical Substances |
| ELINCS | European List of Notified Chemical Substances |
| EmS | Emergency Schedule |
| GHS | "Globally Harmonized System of Classification and Labelling of Chemicals" developed by the United Nations |
| IATA | International Air Transport Association |
| IATA/DGR | Dangerous Goods Regulations (DGR) for the air transport (IATA) |
| ICAO | International Civil Aviation Organization |
| IMDG | International Maritime Dangerous Goods Code |
| index No | the Index number is the identification code given to the substance in Part 3 of Annex VI to Regulation (EC) No 1272/2008 |
| IOELV | indicative occupational exposure limit value |
| MARPOL | International Convention for the Prevention of Pollution from Ships (abbr. of "Marine Pollutant") |
| NLP | No-Longer Polymer |
| PBT | Persistent, Bioaccumulative and Toxic |
| PNEC | Predicted No-Effect Concentration |
| ppm | parts per million |
| REACH | Registration, Evaluation, Authorisation and Restriction of Chemicals |
| RID | Règlement concernant le transport International ferroviaire des marchandises Dangereuses (Regulations concerning the International carriage of Dangerous goods by Rail) |
| STEL | short-term exposure limit |
| SVHC | Substance of Very High Concern |
| TWA | time-weighted average |
| VOC | Volatile Organic Compounds |
| vPvB | very Persistent and very Bioaccumulative |
| WEL | workplace exposure limit |

### Key literature references and sources for data

- Regulation (EC) No. 1907/2006 (REACH), amended by 2015/830/EU
- Regulation (EC) No. 1272/2008 (CLP, EU GHS)
- Dangerous Goods Regulations (DGR) for the air transport (IATA)
- International Maritime Dangerous Goods Code (IMDG)

### List of relevant phrases (code and full text as stated in chapter 2 and 3)

| Code | Text |
|---|---|
| H226 | flammable liquid and vapour |
| H304 | may be fatal if swallowed and enters airways |
| H312 | harmful in contact with skin |
| H315 | causes skin irritation |
| H319 | causes serious eye irritation |
| H332 | harmful if inhaled |
| H335 | may cause respiratory irritation |
| H373 | may cause damage to organs (respiratory system, nervous system) through prolonged or repeated exposure |

### Disclaimer

The above information describes exclusively the safety requirements of the product and is based on our present-day knowledge. The information is intended to give you advice about the safe handling of the product named in this safety data sheet, for storage, processing, transport and disposal. The information cannot be transferred to other products. In the case of mixing the product with other products or in the case of processing, the information on this safety data sheet is not necessarily valid for the new made-up material.